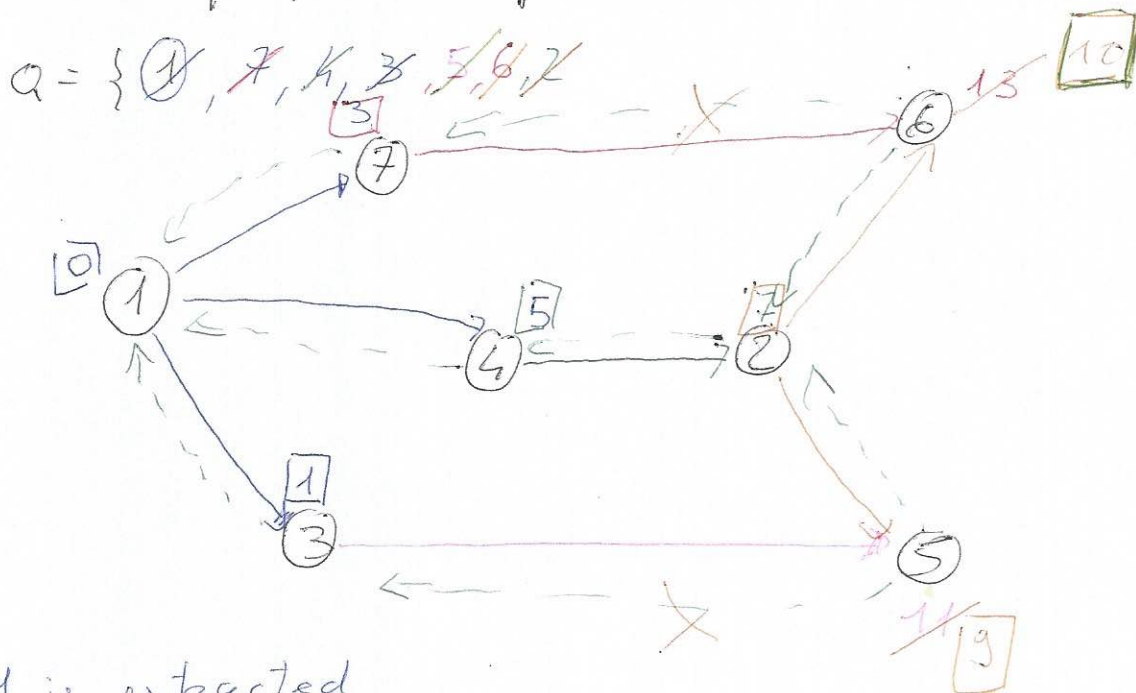


The graph is not acyclic, the costs are 20 Ex 1 (1)
 $\Rightarrow$ the most efficient algorithm is Dijkstra
 no preprocessing is needed



1 is extracted

3 is extracted

7 is extracted

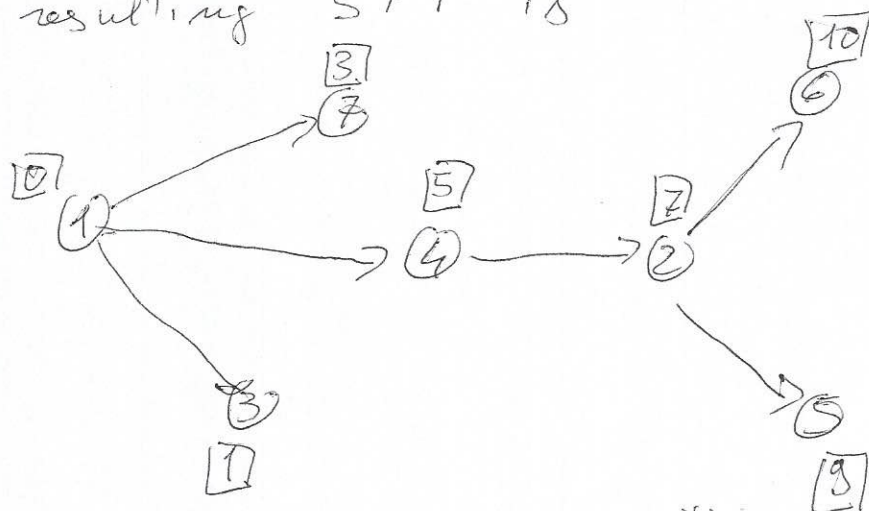
4 is extracted

2 is extracted

5 is extracted

6 is extracted

The resulting SPT is



In order to maintain the optimality of the tree
 the length of $(4,2)$ must be such that all the arcs (i,j)
 satisfy $d[i] + c_{ij} \geq d[j]$ even when the labels of the
 subtree rooted in 2 are updated.

In particular if the length of c_{42} is $2+d$ (2)
and the tree remains unchanged

$d[6]$ will be $10+d$

$d[5]$ will be $9+d$

the only arcs whose feasibility can be affected by the change are $(7,6)$ and $(3,5)$ thus

$$d[7] + c_{76} \geq d[6]$$

$$3 + 10 \geq 10 + d \quad \Rightarrow \quad d \leq 3$$

$$d[3] + c_{35} \geq d[5]$$

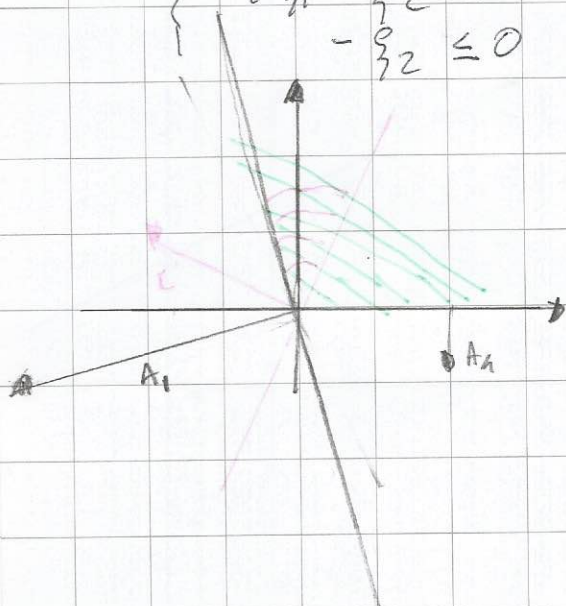
$$1 + 10 \geq 9 + d \quad \Rightarrow \quad d \leq 2$$

In order to maintain the SPT unchanged the length of $(4,2) \leq 2+d$

$$\bar{x} = \begin{bmatrix} -1/2 \\ 0 \end{bmatrix}$$

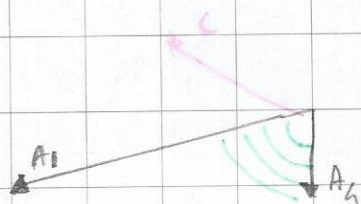
$$I(\bar{x}) = \{1, 4\}$$

$$\text{RP} \quad \begin{cases} -2x_1 + x_2 > 0 & (1) \\ -x_1 - x_2 \leq 0 & (2) \\ -x_2 \leq 0 & (3) \end{cases}$$



has solution

$$\text{RD} \quad \begin{cases} -4u_1 = -2 \\ -u_1 - u_2 = 1 \\ u_1, u_2 \geq 0 \end{cases}$$



has no solution

$$q' = \begin{bmatrix} -1 \\ 0 \end{bmatrix}$$

satisfies (1) $+2 > 0$

satisfies (3) $0 \leq 0$

does not satisfy (2) $4 \neq 0$

it is not a feasible direction though it is a growing one

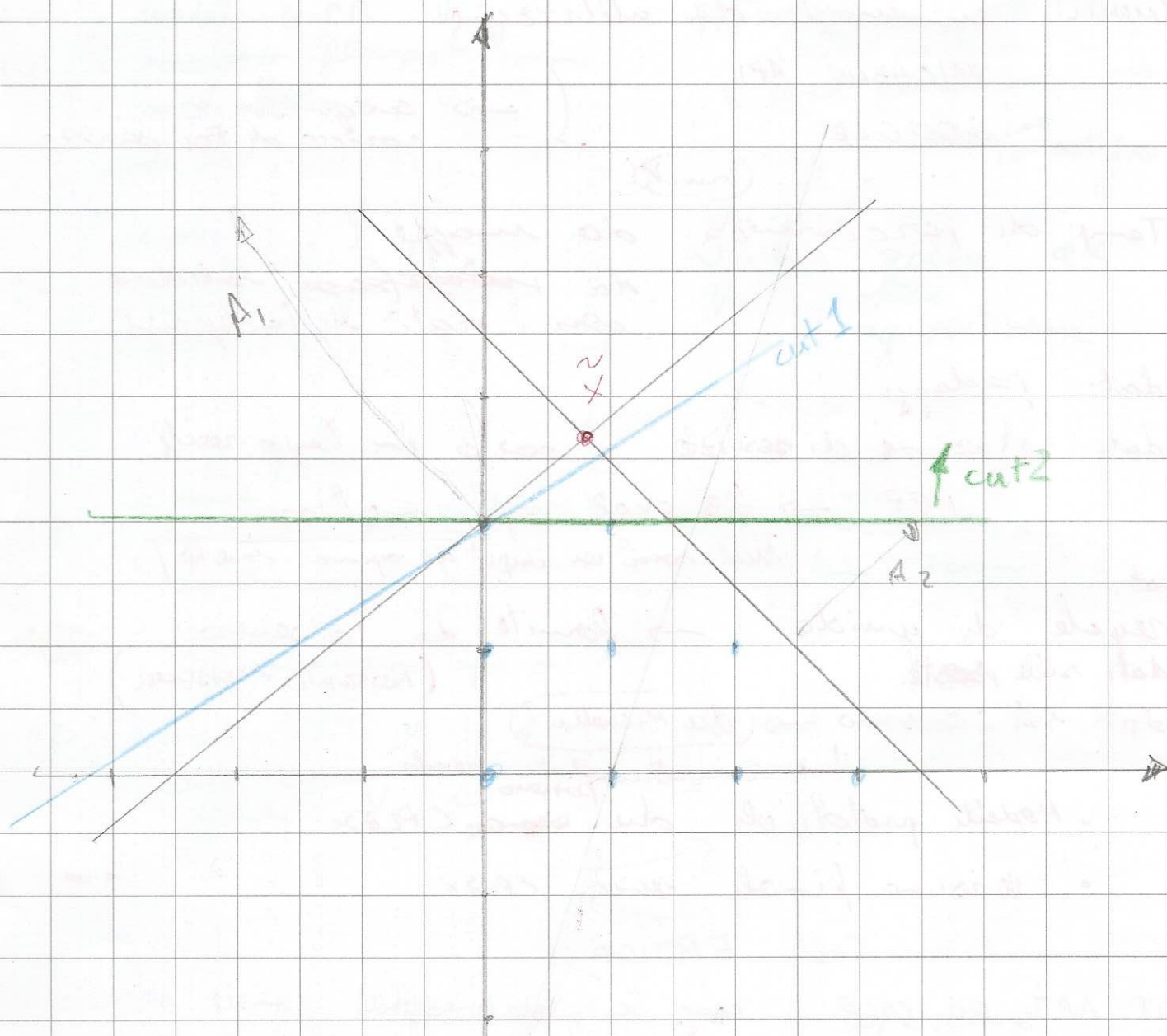
$$q'' = \begin{bmatrix} -1 \\ 4 \end{bmatrix}$$

satisfies (1) $6 > 0$

satisfies (2) $0 \leq 0$

satisfies (3) $-4 \leq 0$

it is a growing feasible direction



$$\max x_1 + 2x_2$$

$$-4x_1 + 5x_2 \leq 10$$

$$2x_1 + 2x_2 \leq 7$$

$$x_1, x_2 \geq 0$$

$$\max x_1 - 2x_2$$

$$-4x_1 + 5x_2 + x_3 = 10$$

$$2x_1 + 2x_2 + x_4 = 7$$

$$x_1, x_2, x_3, x_4 \geq 0$$

(2)

$$A_B = \begin{bmatrix} -4 & 5 \\ 2 & 2 \end{bmatrix}$$

$$A_B^{-1} = \begin{bmatrix} -2/18 & 5/18 \\ 2/18 & 4/18 \end{bmatrix}$$

$$A_N = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$$

$$A_B^{-1} A_N = \begin{bmatrix} -2/18 & 5/18 \\ 2/18 & 4/18 \end{bmatrix}$$

$$A_B^{-1} b = \begin{bmatrix} 15/18 \\ 48/18 \end{bmatrix} = \tilde{x}$$

$$\begin{array}{c} x_1 \\ x_2 \end{array} \begin{array}{cc} x_3 & x_4 \\ \begin{bmatrix} -2/18 & 5/18 \\ 2/18 & 4/18 \end{bmatrix} \end{array} \leq \begin{bmatrix} 15/18 \\ 48/18 \end{bmatrix}$$

Cut 1 : $x_1 - x_3 + 0 x_4 \leq 0$

Cut 2 : $x_2 + 0 x_3 + 0 x_4 \leq 2$

$x_3 = 10 + 4x_1 - 5x_2$

$x_1 - 10 - 4x_1 + 5x_2 \leq 0 \quad -3x_1 + 5x_2 \leq 10$

ICOOK



```
set I := 1..10;
set J := 7..30;
var x{I,J};
param b{J};

s.t. c1 {j in J}:
    sum{i in I: i <= b[j]}x[i,j] <= 5;
```

$$y_{ij} = \begin{cases} 1 & \text{zone } j \text{ served by an antenna in site } i \\ 0 & \text{otherwise} \end{cases}$$

$$\max \sum_{i \in S} \sum_{j \in Z} y_{ij}$$

$$\sum_{i \in S} y_{ij} \leq 1 \quad \forall j \in Z$$

$$[y_{ij} = 0 \quad \forall i \in S \quad \forall j \in Z : p_{ij} < \Delta]$$

or, in alternative

$$p_{ij} \geq y_{ij} \Delta$$

Variant

$$x_i = \begin{cases} 1 & \text{antenna installed in } i \\ 0 & \text{otherwise} \end{cases}$$

$$y_{ij} \leq x_i \quad \forall i \in S \quad \forall j \in Z$$

$$\sum_{h \in S} p_{hj} x_h - p_{ij} y_{ij} \leq \delta + M(1 - y_{ij}) \quad \forall i \in S \quad \forall j \in Z$$

Dual

$$\max 8y_A + 12y_B + y_C$$

$$2y_A + 2y_B \leq 4$$

$$y_A + 3y_B + y_C \leq 3$$

$$y_A, y_B, y_C \geq 0$$

5 equations

$$y_A (2x_1 + x_2 - 8) = 0$$

$$y_B (2x_1 + 3x_2 - 12) = 0$$

$$y_C (x_2 - 1) = 0$$

$$x_1 (2y_A + 2y_B - 4) = 0$$

$$x_2 (y_A + 3y_B + y_C) = 0$$

$$\bar{x} = \begin{bmatrix} 3 \\ 2 \end{bmatrix} \quad x_2 > 1 \Rightarrow y_C = 0$$

$$x_1, x_2 > 0 \Rightarrow y_A = 3/2 \quad y_B = 1/2 \quad \geq 0$$

$$\bar{y} = \left[\frac{3}{2}, \frac{1}{2}, 0 \right] \text{ optimal dual solution}$$

The new feeds 3 and 4 introduce 2 new variables x_3 and x_4 in the primal and the following constraints in the dual

$$3) \quad 4y_B + 2y_C \leq 2$$

$$4) \quad 2y_A + 2y_C \leq 2$$

The dual solution $\bar{y}$ violates the constraint 4)

$\Rightarrow x_4$ is worth introducing in the primal while it does not violate constraint 3)

$\Rightarrow x_3$ is not worth introducing